R16

Code No: 132AC

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year II Semester Examinations, January/February - 2024
MATHEMATICS-III
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define a discrete random variable. [2]

b) If X is a continuous random variable with p.d.f $f(x) = \begin{cases} ax, & 0 \le x \le 1 \\ a \text{ in } & 1 \le x \le 2 \\ 0 & elsewhere \end{cases}$. Find 'a'. [3]

c) Write the formulae to find the standard error of the i) sample mean ii) Difference between the means. [2]

d) A random sample of size 32 is taken from a population whose variance is 8 with probability 0.99, find the maximum error. [3]

e) Define type 1 and type 2 errors. [2]

f) In a random sample of 100 packages shipped by air freight 13 had some damage. Find the standard error of proportions. [3]

g) Find the two values between which the root of $x = \frac{1}{2} + \sin x$ lies. [2]

h) Write the normal equations to fit $y = ax^b$ [3]

i) Write Simpson's 1/3rd and 3/8th formulae to evaluate $\int_a^b y dx$ [2]

j) If $\frac{dy}{dx} = x + y + xy$, h= 0.5, y(0) = 1 then find y(0.5) and y(1) Euler's method. [3]

PART-B

(50 Marks)

2.a) If the chance that any of the 10 telephone lines is busy at an instant is 0.2. What is the most probable number of busy lines and what is the probability of this number.

b) Assume the mean height of soldier to be 68 inches with a variance of 100 inches. How many soldiers in a regiment of 10,000 would you expect to be above 70 inches? [5+5]

OR

3.a) A coin is tossed four times. X is the number of heads. Write the distribution of X. Find: i) Mean ii) variance.

b) 1000 students had written an examination the mean of test is 35 and standard deviation is 5. Assuming the distribution to be normal find**,** how many students marks lie between 25 and 40. [5+5]

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4. Two horses A and B were tested according to the time (in seconds) to run a particular track with the following results.

Horse A	28	30	32	33	35	29	34
Horse B	29	30	30	24	27	29	

Test whether the two horses have the same running capacity. [10]

OR

5. Given below is the number of male births in 1000 families having five children

Male Children	0	1	2	3	4	5
No. of Families	40	300	250	200	30	180

Test whether the given data is consistent with the hypothesis that the binomial law holds if the chance of a male birth is equal to that of female birth. [10]

6. In a city A 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant at 0.05 level of significance? [10]

OR

7.a) A sample of 400 items is taken from a population whose standard deviation is 10. The mean of the sample is 40. Test whether the sample has come from a population with mean 38.

b) The standard deviations of two samples are 8 and 12. Samples sizes are 200 and 100. Find the standard error of the difference between the means and also find the confidence interval at 0.05 level. Means of the sample are 60, 50. [5+5]

8. Fit a parabola $y = a + bx + cx^2$ to the following data.

x	1	2	3	4	5	6	7
y	2.3	5.2	9.7	16.5	29.4	35.5	54.4

[10]

OR

9. Solve the system of equations using Gauss Seidal method

x + 5y + 2z = 7, 7x - y + z = 2, x + 2y + 5z = 9 [10]

10. Given that $\frac{dy}{dx} = e^x - y$ and y(0)=1. Compute y(0.1) and y(0.2) using Picard's method. [10]

OR

11. Use Runge Kutta fourth order method to evaluate y at x= 0.1 and 0.2 given that $\frac{dy}{dx} = x + y$, y(0) = 1 [10]

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